Linear Algebra Done Wrong (Sergei Treil)

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1. Find a basis in the space of 3×2 matrices $M_{3\times 2}$.

   We can do six unit vectors as follows:
$$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

2. True or false:
   (a) Any set containing a zero vector is linearly dependent.
   true, there is a nontrivial linear combination if we let the coefficient of the zero vector be 1 and everything else be 0.
   (b) A basis must contain $\mathbf{0}$
   false, a basis cannot contain $\mathbf{0}$, as then it is not linearly independent.
   (c) subsets of linearly dependent sets are linearly dependent.
   false, take $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\}$, which is a linearly dependent set, but the subset $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ is not linearly dependent.
   (d) subsets of linearly independent sets are linearly independent.
   true. If there were a linearly dependent subset A of a linearly independent set B, that means there is a nontrivial linear combination of vectors in A that sums to $\mathbf{0}$. However, since A is a subset of B, there is a nontrivial linear combination of vectors in B that yields $\mathbf{0}$ (as we just let the coefficients of elements not in A be 0 while the coefficients of elements in both is 1).
   (e) If $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n = \mathbf{0}$ then all scalars α_k are zero.
   false. Let $n = 2$ and $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$. Then, $\alpha_1 = -2$ and $\alpha_2 = 1$ provides a nontrivial solution to $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 = \mathbf{0}$.

3. Recall, that a matrix is called *symmetric* if $A^T = A$. Write down a basis in the space of *symmetric* 2×2 (there are many possible answers). How many elements are in the basis?
   If a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is symmetric, then it is equal to $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$, so we must have $b = c$.

   We must be able to make any combination of a, $b = c$, and d, so we will need three matrices:
$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

4. Write down a basis for the space of
   (a) 3×3 symmetric matrices
   For a matrix of the form $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, its transpose is $\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$. We see here that for the matrix

to be symmetric then it is necessary and sufficient for $b = d$, $c = g$, and $f = h$ to hold.

Thus, we can find a unique 3×3 symmetric matrix for all all tuples (a, b, c, e, f, i), meaning we need 6 matrices in our basis:

$$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

(b) $n \times n$ symmetric matrices

For a matrix of the form $\begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \dots & a_{n,n} \end{bmatrix}$ to be symmetric, $a_{i,j} = a_{j,i}$ must hold for all $1 \leq i, j \leq n$.

Giving a unique value to all $a_{i,j}$ such that $i \leq j$ uniquely defines an $n \times n$ symmetric matrix, and all such matrices are representable this way. The number of such pairs (i, j) is $\frac{n(n+1)}{2}$, as there are n possible values for i, and if for j we say there are $n+1$ choices (a number from $1 \dots n$ or the same value as i) then we will have counted each pair twice, hence the dividing by 2.

Let $E_{i,j}$ be an $n \times n$ matrix entirely consisting of 0's except for $E_{i,j_{i,j}}$, which equals 1. Our basis is:

$$B = \{E_{i,j} \mid i \leq j\}$$

(c) $n \times n$ *antisymmetric* ($A^T = -A$) matrices.

For $A^T = -A$ to hold, it must be the case that $a_{i,j} = -a_{j,i}$ for all $1 \leq i, j \leq n$. Note that if $i = j$, then $a_{i,j} = 0$. Thus, we have $\frac{n(n-1)}{2}$ pairs (i, j) which can take on any value, since for every (i, j) such that $i < j$, we can select a value for $a_{i,j}$.

We must define matrices $C_{i,j}$ such that all entries are 0, except $C_{i,j_{i,j}} = 1$ and $C_{i,j_{j,i}} = -1$. With this, our answer is:

$$B = \{C_{i,j} \mid i < j\}$$

(d) Let a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1}$ such that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent. **Hint:** Take for $\mathbf{v}_{r+1}$ any vector that cannot be represented as a linear combination $\sum_{k=1}^{r} \alpha_k \mathbf{v}_k$ and show that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

We will do as the hint says. Since $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ is not generating, there is some vector $\mathbf{v}_{r+1}$ which is not a linear combination of $\mathbf{v}_1 \dots \mathbf{v}_r$.

Now it remains to prove that $\mathbf{v}_1 \dots \mathbf{v}_{r+1}$ is linearly independent. We will now show that there is no linear combination of $\mathbf{v}_1 \dots \mathbf{v}_{r+1}$ except the trivial solution that produces $\mathbf{0}$.

We will use a proof by contradiction. Assume this were the case, and there exists some $\alpha_1 \dots \alpha_{r+1}$ such that $\sum_{k=1}^{r+1} \alpha_k \mathbf{v}_k = \mathbf{0}$ where there exists k such that $\alpha_k \neq 0$. If $\alpha_{r+1} = 0$, then it would mean there is a trivial linear combination of $\mathbf{v}_1 \dots \mathbf{v}_r$ that produces $\mathbf{0}$, contradicting the fact that it is a linearly independent set. Thus, we must have $\alpha_{r+1} \neq 0$.

Furthermore, since $\mathbf{v}_{r+1} \neq \mathbf{0}$ (as $\mathbf{0}$ cannot be part of a linearly independent set), there must exist $k < r+1$ such that $\alpha_k \neq 0$, as otherwise $\sum_{k=0}^{r+1} \alpha_k \mathbf{v}_k = \alpha_{r+1}\mathbf{v}_{r+1} \neq 0$. Thus, we have:

$$\sum_{k=1}^{r+1} \alpha_k \mathbf{v}_k = \mathbf{0}$$

$$\sum_{k=1}^{r} \alpha_k \mathbf{v}_k = -\alpha_{r+1}\mathbf{v}_{r+1}$$

$$\sum_{k=1}^{r} -\frac{\alpha_k}{\alpha_{r+1}}\mathbf{v}_k = \mathbf{v}_{r+1}$$

As such, the fact that we have found a linear combination of $\mathbf{v}_1 \dots \mathbf{v}_r$ that produces $\mathbf{v}_{r+1}$ contradicts our original assumption of $\mathbf{v}_{r+1}$.

We have shown through contradiction that if $\mathbf{v}_{r+1}$ (which must exist) cannot be represented as a linear combination of $\mathbf{v}_1 \dots \mathbf{v}_r$, the set $\mathbf{v}_1 \dots \mathbf{v}_{r+1}$ is linearly independent.

5. Is it possible that vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent but the vectors $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$, $\mathbf{w}_3 = \mathbf{v}_3 + \mathbf{v}_1$ are linearly independent?

If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, then without loss of generality let $\mathbf{v}_3 = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2$, as for a set of vectors to be linearly dependent, at least one vector is a linear combination of the others.

It must be the case that there are some coefficients $\alpha_1, \alpha_2, \alpha_3$ such that $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = \mathbf{0}$.

We can observe that $\mathbf{v}_1 = \frac{1}{2}(\mathbf{w}_1 + \mathbf{w}_3 - \mathbf{w}_2)$, so $\alpha_1\mathbf{v}_1 = \frac{\alpha_1}{2}(\mathbf{w}_1 + \mathbf{w}_3 - \mathbf{w}_2)$. Similarly, $\alpha_2\mathbf{v}_2 = \frac{\alpha_2}{2}(\mathbf{w}_1 + \mathbf{w}_2 - \mathbf{w}_3)$, and $\alpha_3\mathbf{v}_3 = \frac{\alpha_3}{2}(\mathbf{w}_2 + \mathbf{w}_3 - \mathbf{w}_1)$. If we sum them all together, we get

$$\frac{\alpha_1}{2}(\mathbf{w}_1 + \mathbf{w}_3 - \mathbf{w}_2) + \frac{\alpha_2}{2}(\mathbf{w}_1 + \mathbf{w}_2 - \mathbf{w}_3) + \frac{\alpha_3}{2}(\mathbf{w}_2 + \mathbf{w}_3 - \mathbf{w}_1) = \mathbf{0}$$

because $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3 + \alpha_3\mathbf{v}_3 = \mathbf{0}$. Now all that remains is to ensure this linear combination is nontrivial.

$$(\frac{\alpha_1 + \alpha_2 - \alpha_3}{2})\mathbf{w}_1 + (\frac{\alpha_2 + \alpha_3 - \alpha_1}{2})\mathbf{w}_2 + (\frac{\alpha_1 + \alpha_3 - \alpha_2}{2})\mathbf{w}_3 == \mathbf{0}$$

We want to know if it's possible for for the coefficients to be equal to 0 while $\alpha_1, \alpha_2, \alpha_3$ are not all zero. We have:

$$\alpha_1 + \alpha_2 - \alpha_3 = 0$$

$$\alpha_2 + \alpha_3 - \alpha_1 = 0$$

$$\alpha_1 + \alpha_3 - \alpha_2 = 0$$

We isolate α_3 in the first equation: $\alpha_3 = \alpha_1 + \alpha_2/$. We substitute into the second equation:

$$\alpha_2 + \alpha_1 + \alpha_2 - \alpha_1 = 0$$

$$2\alpha_2 = 0$$

$$\alpha_2 = 0$$

And the third:

$$\alpha_1 + \alpha_1 + \alpha_2 - \alpha_2 = 0$$

$$2\alpha_1 = 0$$

$$\alpha_1 = 0$$

Since $\alpha_1 = 0, \alpha_2 = 0,\ \alpha_3 = 0$. Thus, we conclude that the only solution to the above system is $\alpha_1 = 0, \alpha_2 = 0, \alpha_3 = 0$. Since we know at least one of $\alpha_1, \alpha_2, \alpha_3$ isn't zero, then at least one of our new coefficients $\frac{\alpha_1+\alpha_2-\alpha_3}{2}, \frac{\alpha_2+\alpha_3-\alpha_1}{2}, \frac{\alpha_1+\alpha_3-\alpha_2}{2}$ is not zero. Thus, we have found a nontrivial linear combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ to get $\mathbf{0}$, meaning that they cannot be linearly independent.